IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Pedro Mazzarolo Vizeu Pereira
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Outline

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Executive Summary

To address these questions, a comprehensive Data Science pipeline was implemented:

- Data Collection & Wrangling: Data was gathered via SpaceX API and Web Scraping (Wikipedia), followed by thorough cleaning and feature engineering (One-Hot Encoding).
- Exploratory Data Analysis (EDA): Leveraged SQL for statistical insights and Folium for geographical proximity analysis of launch sites.
- Interactive Analytics: Developed a real-time Plotly Dash dashboard to filter and visualize success trends by payload and location.
- Predictive Modeling: Trained and optimized (via GridSearchCV) four classification algorithms: Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors (KNN).

Results:

- Top Performance: KSC LC-39A was identified as the most successful launch site, showing the highest booster recovery rate.
- Payload Insight: Analysis revealed an optimal success "sweet spot" for payloads weighing between 2,000 kg and 6,000 kg.
- Model Accuracy: All classification models achieved a consistent 83.33% accuracy on the test set, with Logistic Regression selected as the most efficient model for this task.
- Technological Maturity: The data confirms a clear learning curve, with newer rocket iterations achieving near-perfect reliability.

Introduction

Project Background and Context

SpaceX has revolutionized the aerospace industry by making space travel significantly more affordable. By accurately predicting the success of a booster landing, we can determine the actual cost of a mission and provide a competitive edge in the market.

Problems to Solve (Research Questions):

- What geographical and technical factors (such as payload mass and orbit type) most influence a successful landing?
- How has the success rate evolved across different generations of the Falcon 9 booster?
- Can a robust Machine Learning model be built to forecast mission outcomes with high accuracy using publicly available data?

Section 1

Methodology

Methodology

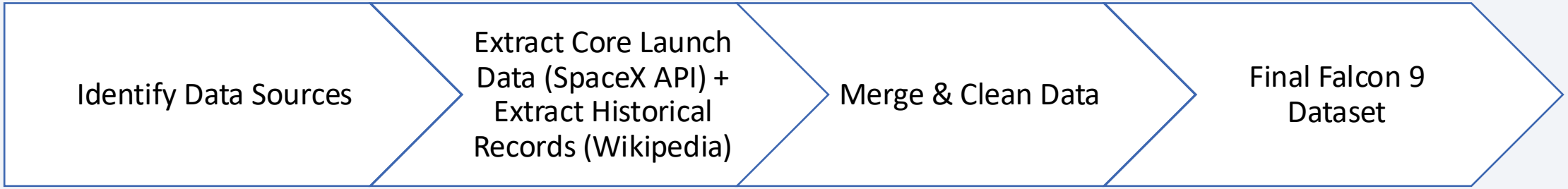
Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

To build a robust predictive model, historical launch data was gathered from two primary sources. We utilized the SpaceX REST API to retrieve core flight telemetry and mission specifics, while Wikipedia web scraping provided supplementary launch records. This dual-source approach ensured a comprehensive and accurate dataset focused exclusively on Falcon 9 launches.

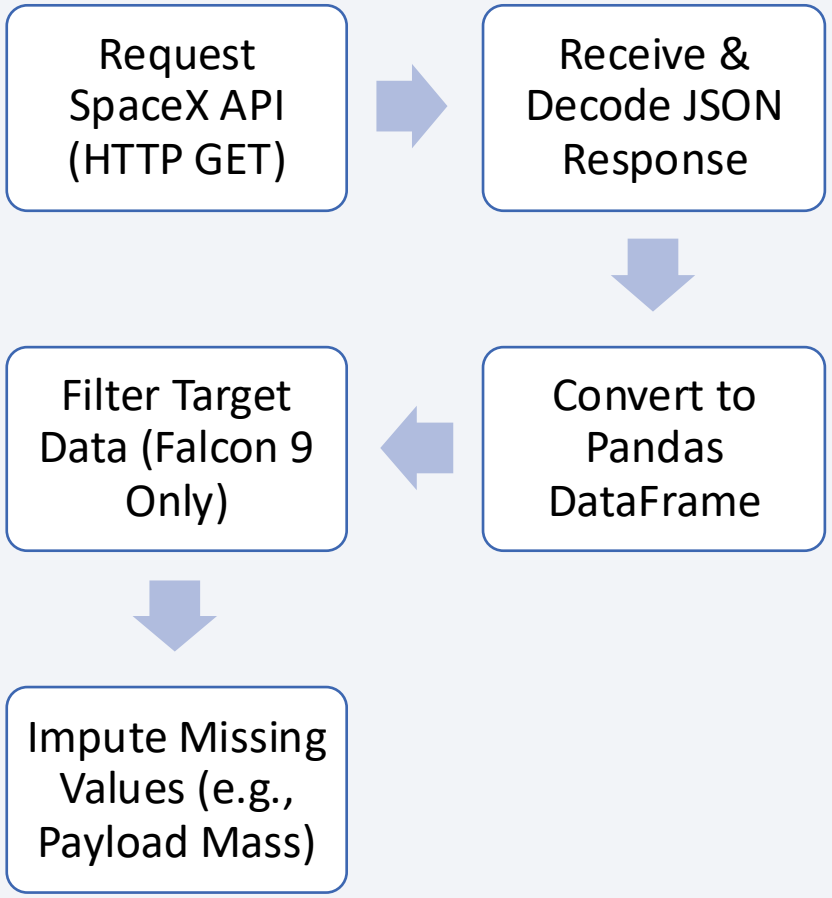
- Sourced historical launch records from multiple digital platforms.
- Combined automated REST API requests and HTML web scraping techniques.
- Unified and filtered the extracted data to focus exclusively on Falcon 9 launches.
- Created a robust, structured dataset for predictive modeling.



Data Collection – SpaceX API

The primary dataset was collected by querying the SpaceX REST API. The resulting JSON response was parsed and converted into a structured Pandas DataFrame. Key data cleaning steps included filtering the records to isolate Falcon 9 missions and handling missing values, such as calculating the mean to impute missing Payload Mass, ensuring overall data integrity.

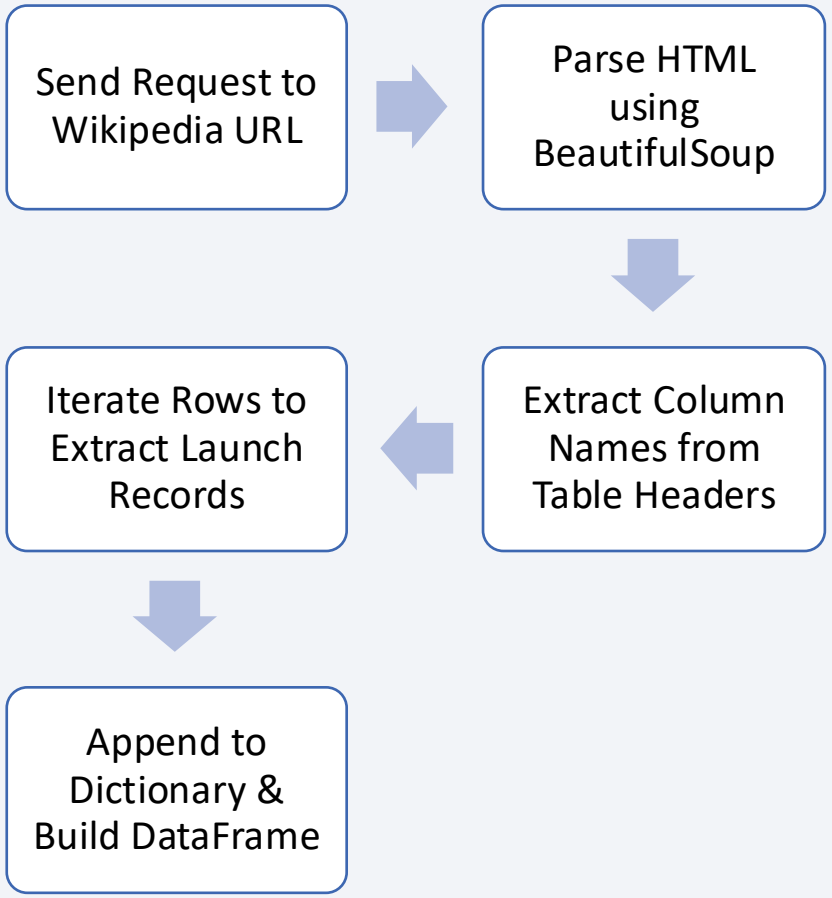
- Executed HTTP GET requests to the SpaceX REST API
- Parsed complex JSON responses into structured Pandas DataFrames.
- Filtered the dataset to include only "Falcon 9" rocket launches.
- Handled missing data, including the calculation of mean values for payload mass imputation.



Data Collection - Scraping

To complement the API data, additional Falcon 9 launch records were extracted from Wikipedia. Using the Python BeautifulSoup library, we parsed the HTML structure of the page, specifically targeting the launch tables. By extracting the table headers and iterating through the data rows, we successfully transformed raw HTML web data into a structured Pandas DataFrame.

- HTTP GET requests to fetch the Falcon 9 Wikipedia page HTML.
- Created a BeautifulSoup object to parse the HTML DOM structure.
- Extracted table headers to dynamically define DataFrame column names.
- Iterated through HTML table rows to extract individual launch records.
- Constructed a dictionary and converted it into a cleaned Pandas DataFrame.

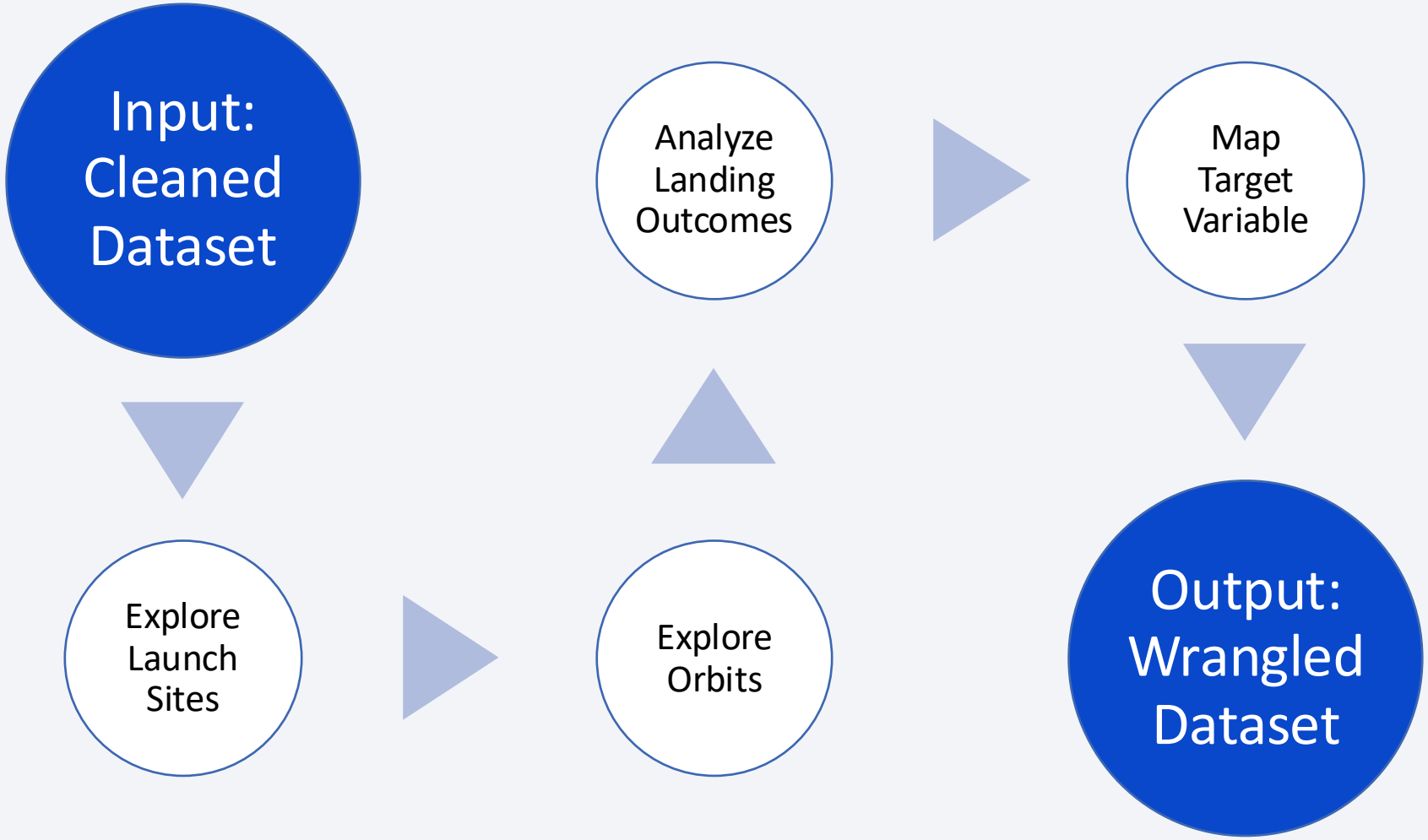


Data Wrangling

The data wrangling process began with an exploratory analysis of categorical variables to understand the dataset's distribution. First, I determined the number of launches per launch site and calculated the occurrences of each orbit type. Following this initial exploration, I focused on the landing outcomes. The Outcome column contained various string values which I analyzed and grouped to isolate all unsuccessful landing attempts into a bad_outcomes set. Finally, I created a new binary target variable named Class, mapping successful landings to 1 and unsuccessful ones to 0, successfully preparing the dataset for future predictive modeling.

- **Launch Site Analysis** Determined the number of launches across different sites (e.g., CCAFS SLC 40, KSC LC 39A).
- **Orbit Distribution:** Calculated the number and occurrence of each specific orbit type.
- **Outcome Exploration:** Analyzed the distribution of categorical landing outcomes to identify success vs. failure patterns.
- **Binary Classification & Labeling:** Grouped failed attempts and mapped the string outcomes into a numerical binary target variable (`Class`: 1 = Success, 0 = Failure).

Data Wrangling



EDA with Data Visualization

During the Exploratory Data Analysis (EDA) phase, several charts were plotted to uncover hidden patterns and correlations within the dataset. Categorical scatter plots were utilized to visualize the distribution of landing outcomes across different flight numbers, payload masses, launch sites, and orbit types. A bar chart was employed to calculate and compare the mean success rate of each specific orbit. Finally, a line chart was plotted to track the historical success rate over time. These visualizations were crucial to understand feature relationships and determine which variables most heavily impact the first-stage landing success, laying the groundwork for feature selection in Machine Learning.

Categorical Scatter Plots

What: Plotted combinations like Flight Number vs. Launch Site, Payload Mass vs. Launch Site, and Payload Mass vs. Orbit, using colors to separate Success (1) from Failure (0).
Why: To see if certain launch sites are used only for heavy payloads, or if higher flight numbers (more recent launches) show a denser cluster of successful landings.

Bar Chart

What: Plotted Orbit Type vs. Success Rate (calculating the mean of the Class column).
Why: Bar charts are the best tool for comparing aggregated metrics across categories. It instantly showed which orbits (like SSO, HEO, ES-L1) have 100% success rates, and which ones are more challenging (like GTO or SO).

Line Chart

What: Plotted Year/Date vs. Average Success Rate.
Why: Line charts are standard for time-series data. It was used to visually prove the hypothesis that as SpaceX gained experience over the years, their success rate drastically improved.

EDA with SQL

- **Launch Site Identification:** Retrieved unique launch sites and utilized string pattern matching to isolate historical records from Cape Canaveral facilities.
- **Payload Statistics:** Performed aggregate calculations to determine the total payload mass delivered for NASA (CRS) and the average mass for specific booster versions like F9 v1.1.
- **Operational Milestones:** Pinpointed the exact date of the first successful ground pad landing and identified specific boosters that achieved successful drone ship landings within defined weight constraints.
- **Mission Success Metrics:** Quantified the total frequency of successful and failed mission outcomes to evaluate overall program reliability.
- **Advanced Data Retrieval:** Implemented subqueries to extract all booster versions responsible for carrying the maximum recorded payload mass in the dataset.
- **Temporal Ranking:** Analyzed failure trends for 2015 and ranked the frequency of all landing outcomes within a specific historical window (2010–2017).

Build an Interactive Map with Folium

Folium Map Objects & Implementation:

- Circles: Added to each launch site with a 1km radius to visually highlight the specific facility area and provide immediate site identification through popups.
- Markers: Utilized to pinpoint exact launch coordinates and specific landing locations for mission outcome visualization.
- Marker Clusters: Implemented to group high-density launch data at each site, preventing visual clutter while allowing for a clear success-to-failure ratio analysis using color-coded icons (Green/Red).
- Mouse Position Tool: Integrated a coordinate tracker to allow users to find the exact latitude and longitude of any point of interest on the map for further distance analysis.
- PolyLines: Drew paths between launch pads and nearby infrastructure to visually represent the calculated shortest distances (Haversine formula).

Objectives & Rationale:

- To verify the proximity of launch sites to the equator (fuel efficiency) and the coastline (public safety).
- To analyze logistical efficiency by measuring distances to essential transport infrastructure like highways and railways.
- To assess site isolation from densely populated cities to minimize acoustic and safety risks.

Build a Dashboard with Plotly Dash

Dashboard Components & Graphs:

- **Launch Site Dropdown Menu:** A search-enabled dropdown that allows the user to filter the dashboard by specific launch sites (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40) or view the "All Sites" aggregate.
- **Dynamic Pie Chart:** Visualizes the success distribution. When "All Sites" is selected, it shows the total successful launches contributed by each site. When a specific site is selected, it displays the success-to-failure ratio for that facility.
- **Payload Range Slider:** An interactive slider ranging from 0 to 10,000 kg that allows users to filter the scatter plot data based on mission-specific payload weights.
- **Payload vs. Outcome Scatter Plot:** A multi-dimensional graph that plots Payload Mass against Launch Outcome (Class 0 or 1), using color-coded markers to represent different Booster Version Categories.

Objectives & Rationale:

- **Comparative Analysis:** The dropdown and pie chart were added to quickly identify which launch sites are the most productive and which have the highest operational reliability.
- **Trend Identification:** The scatter plot was implemented to discover if certain payload ranges are more "at-risk" for landing failures and to visually confirm the superior performance of newer booster versions (like Block 5) compared to earlier iterations.
- **User Empowerment:** These interactions allow stakeholders to "slice and dice" the data without writing code, making complex mission trends accessible through visual interaction.

Predictive Analysis (Classification)

Model Development Process

1. **Data Preprocessing & Standardization:** Features were standardized using StandardScaler to ensure that all input variables had a mean of zero and a variance of one, which is critical for the performance of algorithms like SVM and KNN.
2. **Data Splitting:** The dataset was divided into training and testing sets (80/20 split) to allow for unbiased evaluation of the models on unseen data.
3. **Hyperparameter Optimization:** Utilized GridSearchCV with 10-fold cross-validation for each algorithm. This "improvement" step allowed for the automatic selection of the best parameters (such as C, kernels, max_depth, and n_neighbors).
4. **Model Comparison:** Four classification algorithms were trained and evaluated:
    1. Logistic Regression
    2. Support Vector Machine (SVM)
    3. Decision Tree
    4. K-Nearest Neighbors (KNN)
5. **Performance Evaluation:** Accuracy was measured on both the validation (training) set and the test set. Additionally, Confusion Matrices were generated to analyze the distribution of True Positives, True Negatives, False Positives, and False Negatives.
6. **Best Model Identification:** While all models achieved a consistent test accuracy of 83.33%, Logistic Regression was identified as the best performing method due to its optimal balance of predictive power and computational efficiency for this binary classification task.

Results

Exploratory Data Analysis (EDA) Results

- **Geospatial Insights:** All SpaceX launch sites are strategically positioned near the coastline for safety (over-water flight paths) and close to the equator to maximize the Earth's rotational boost for fuel efficiency.
- **Infrastructure Synergy:** Proximity analysis confirmed that launch pads are located within 2km of railways and highways, ensuring efficient logistics for massive booster components, while maintained at a safe distance (~16km) from major population centers like Titusville.
- **SQL Benchmarking:** Identified KSC LC-39A as a key historical site with the highest frequency of successful landings.
  - Calculated the average payload mass for the F9 v1.1 version at 2,928.4 kg.
  - Confirmed the first successful ground landing occurred on 2015-12-22.

Results

Predictive Analysis Results

- **Model Performance:** Four classification models (Logistic Regression, SVM, KNN, and Decision Tree) were rigorously trained and optimized via GridSearchCV.
- **Accuracy Benchmark:** All four models achieved a consistent **Test Accuracy of 83.33%**. This parity suggests that the underlying patterns in the SpaceX dataset are robust and well-captured by standard classification algorithms.
- **Best Model Selection: Logistic Regression** was identified as the optimal model for deployment. It provides a perfect balance of high predictive accuracy and computational simplicity, accurately predicting every successful landing in the test set.
- **Conclusion:** The high accuracy of these models proves that SpaceX's first-stage reuse is not just a feat of engineering, but a highly predictable operational process based on telemetry and mission parameters.

Interactive Analytics Demo (Dashboard)

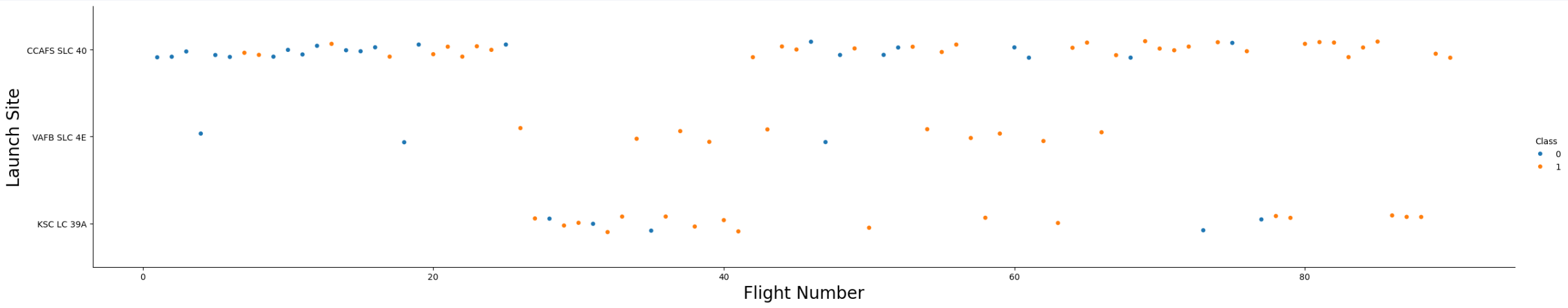
- **Operational Success (Pie Charts):** The interactive dashboard reveals that **KSC LC-39A** is the most reliable facility, boasting a **76.9% success rate**.
- **Payload vs. Outcome (Scatter Plots):**
  - **The "Sweet Spot":** By interacting with the payload range slider, a clear performance window was identified between **2,000 kg and 6,000 kg**, where landing success is most frequent.
  - **Booster Evolution:** The color-coded scatter plot visually demonstrates the technological leap of the **Block 5 (B5)** and **Full Thrust (FT)** versions, which consistently outperform earlier iterations across all payload masses.

Section 2

Insights drawn from EDA

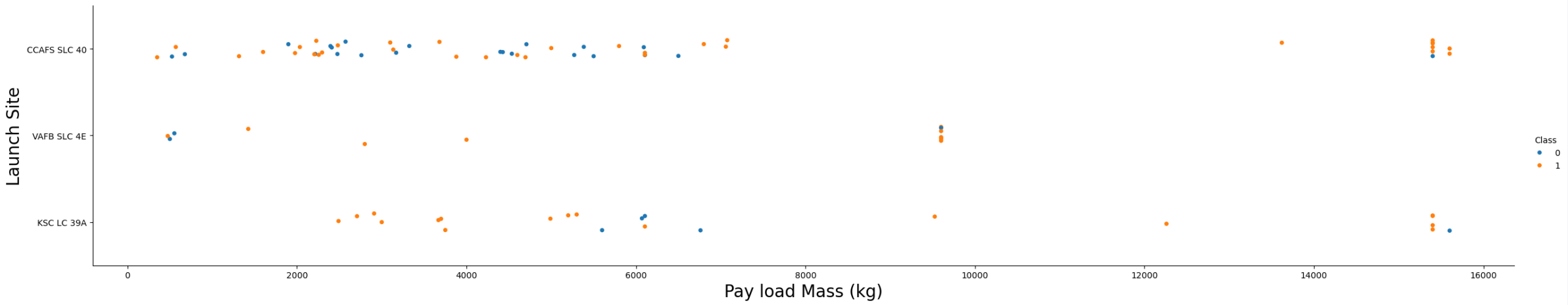
Flight Number vs. Launch Site

The scatter plot of Flight Number vs. Launch Site reveals a clear trend: landing success rates improved significantly over time across all sites. It also highlights that CCAFS SLC 40 is the most frequently used site, while KSC LC 39A, introduced later in the program, exhibits a remarkably high success rate.



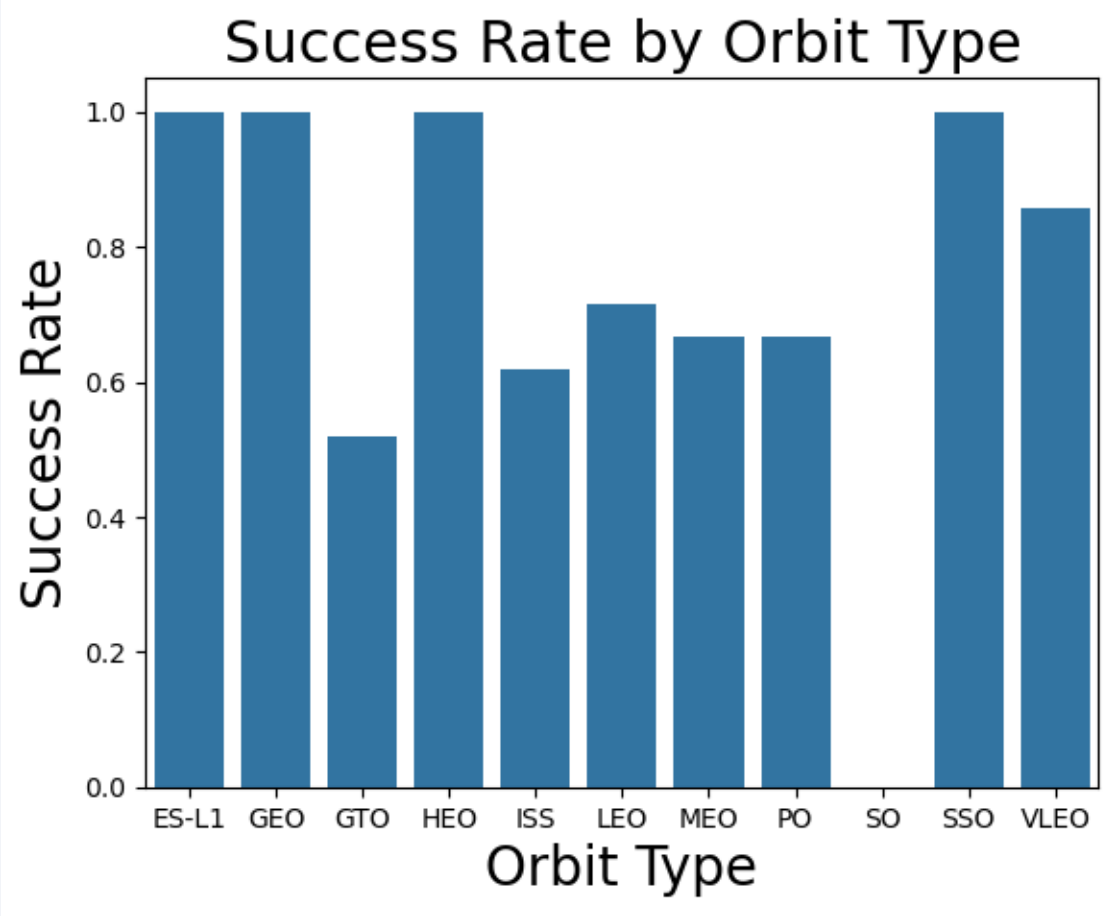
Payload vs. Launch Site

The scatter plot analyzing Payload Mass versus Launch Site reveals distinct operational patterns. Notably, the VAFB SLC 4E site is not utilized for extremely heavy payloads (above 10,000 kg). Furthermore, the CCAFS SLC 40 site handles the majority of launches and shows a remarkably high success rate for heavier payloads (greater than 8,000 kg), whereas lighter payloads at this site experienced mixed outcomes.



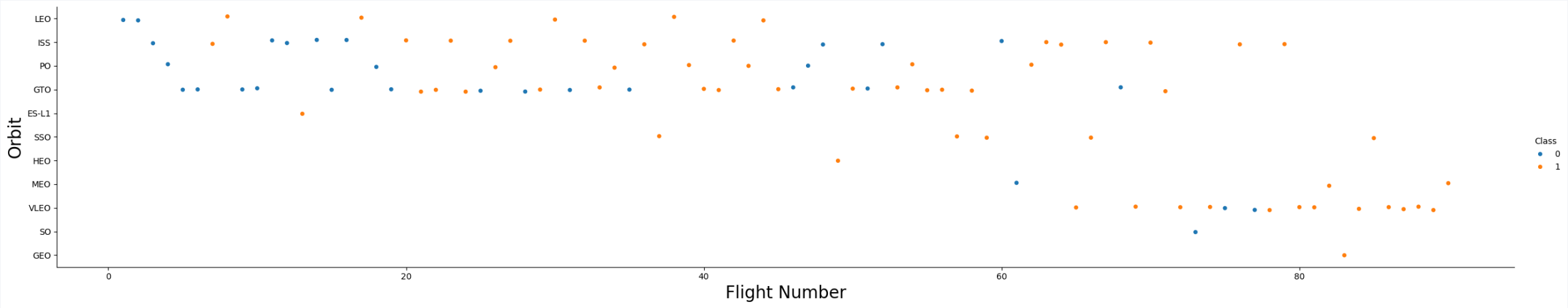
Success Rate vs. Orbit Type

The bar chart illustrating the success rate by orbit type highlights significant variations in landing performance. Orbits such as ES-L1, GEO, HEO, and SSO achieved a flawless 100% success rate. In contrast, GTO missions demonstrated a success rate of approximately 50%, which is consistent with the high-energy requirements of these launches that leave minimal fuel reserves for the booster's return and landing.



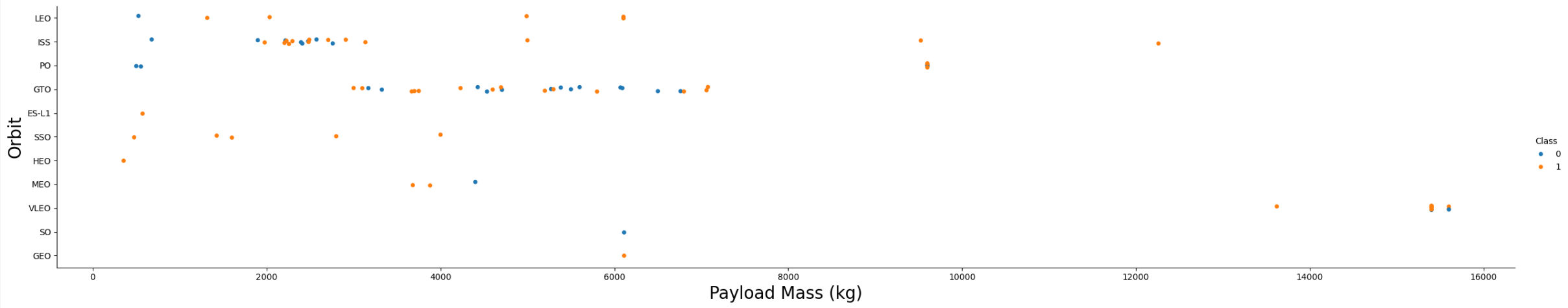
Flight Number vs. Orbit Type

The line chart tracking the launch success yearly trend illustrates a dramatic improvement in SpaceX's first-stage landing capabilities. During the early experimental years (2010-2013), the success rate was near zero. However, from 2014 onwards, there is a sharp and consistent upward trend. By 2020, the success rate stabilized at a highly reliable level of over 80%, clearly demonstrating the maturation and mastery of their reusable rocket technology.



Payload vs. Orbit Type

The scatter plot mapping Flight Number against Orbit type illustrates the progression of SpaceX's capabilities across different mission profiles. LEO and ISS missions show a clear learning curve, with early failures transitioning into consistent successes. GTO missions, while frequent throughout the timeline, remain challenging, displaying a mix of successes and failures even in recent flights due to high energy requirements. Additionally, the dense cluster of successful VLEO missions in later flights highlights the highly reliable deployment phase of the Starlink satellite constellation.



Launch Success Yearly Trend

The scatter plot of Payload Mass versus Orbit reveals a strong relationship between mission parameters and landing success. Notably, extremely heavy payloads (around 15,000 kg) destined for VLEO show a remarkably high success rate, reflecting the highly optimized deployment of the Starlink satellite constellations. In contrast, GTO missions, despite carrying lighter payloads (typically 4,000 - 8,000 kg), experience mixed landing outcomes. This highlights that the high-energy demands of reaching geostationary transfer orbits are a bigger limiting factor for booster recovery than the payload mass itself.

All Launch Site Names

By using the SELECT DISTINCT statement, the query filters out all duplicate entries, revealing that the dataset encompasses only these 4 unique SpaceX launch facilities.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

The LIKE 'CCA%' wildcard operator combined with LIMIT 5 was utilized to filter and extract the first five chronological launch records specifically originating from the Cape Canaveral spaceports.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

The SUM() aggregate function was applied to the payload mass column, filtered by the WHERE clause for the customer "NASA (CRS)", revealing the total mass transported for this specific client.

SUM("PAYLOAD_MASS__KG_")
45596

Average Payload Mass by F9 v1.1

Using the AVG() function filtered for the "F9 v1.1" booster version, this query calculates the mean weight of the payloads carried during that specific rocket generation's lifespan

AVG("PAYLOAD_MASS__KG_")
2928.4

First Successful Ground Landing Date

The MIN() function was applied to the Date column while filtering for a "Success (ground pad)" outcome, effectively pinpointing the exact historical date of SpaceX's first successful terrestrial landing.

MIN("Date")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

This query uses multiple AND operators to filter records that simultaneously meet three conditions: a successful drone ship landing, and a payload mass strictly between 4,000 kg and 6,000 kg.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

By using the GROUP BY clause on the Mission Outcome column alongside the COUNT() function, the query categorizes and counts the absolute number of successful versus failed overall mission objectives.

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

A SQL subquery was employed to first find the absolute maximum payload mass in the dataset using MAX(), and the main query then retrieved all booster versions that matched this exact maximum value.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Using the substr() string manipulation function, the query isolates the year and month from the Date column to find all failed drone ship landings specifically within the year 2015.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

The query utilizes the BETWEEN operator to define a specific date range. It then applies GROUP BY to aggregate the landing outcomes, and ORDER BY DESC to rank them from the most frequent to the least frequent.

Landing_Outcome	Total_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

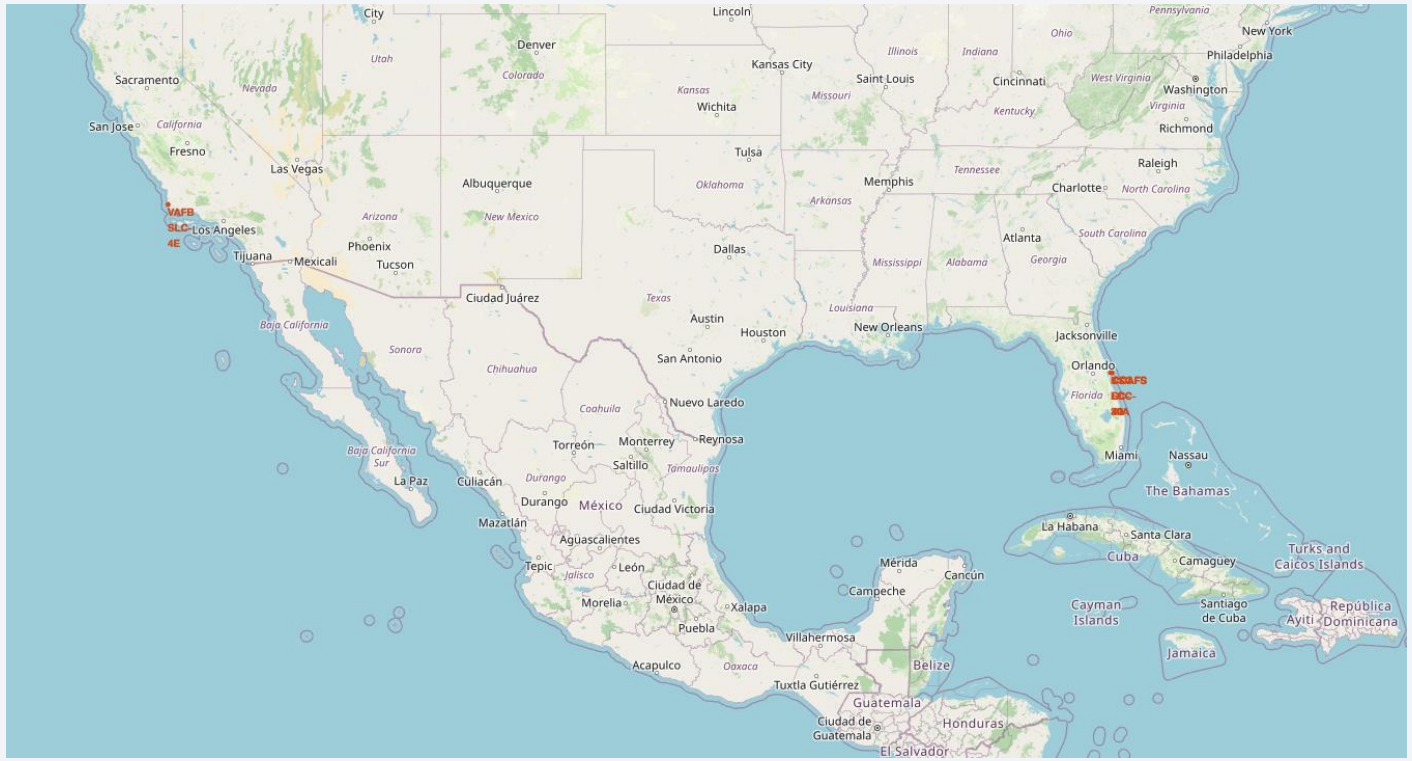
Launch Sites Proximities Analysis

Geographical Analysis of SpaceX Launch Sites

This Folium map displays the geographical distribution of SpaceX's launch facilities. The markers highlight four distinct sites: three clustered on the East Coast in Florida (CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A) and one on the West Coast in California (VAFB SLC-4E). Strategically, all sites are situated directly adjacent to the coastline, ensuring that flight paths are directed over the ocean for maximum safety during launch and stage separation.

Proximity to the Equator: All launch sites are located as close to the Equator as possible. Launching near the Equator provides an extra speed boost from the Earth's rotation, saving fuel and allowing for heavier payloads.

Proximity to the Coast: Every active SpaceX launch site is situated right next to the ocean. This is a critical safety measure; in the event of an explosion or a failed stage separation, debris will fall safely into the water rather than over populated areas.



Landing Success Clusters

Visualizing Outcomes: Using Folium's MarkerCluster plugin, we mapped individual launch outcomes. Green markers represent successful booster landings, while red markers indicate failures.

Site Reliability: The clusters visually confirm that CCAFS SLC-40 has the highest volume of launches, but KSC LC-39A shows a remarkably high ratio of green markers (successes) compared to its total launches.

This map utilizes Folium's MarkerCluster feature to visualize the landing outcomes across all launch sites. Green markers denote successful first-stage landings, while red markers represent failures. The clustered view immediately reveals the volume and performance of each site, highlighting that while CCAFS SLC-40 handled the highest number of flights, KSC LC-39A achieved a remarkably high success ratio.

Distances to Proximities (Safety & Logistics)

Distances:
Using the Haversine formula and Folium PolyLine, distances were calculated from launch sites to nearby points of interest.

Strategic Isolation:
Launch sites are deliberately kept several kilometers away from major cities to minimize acoustic shockwaves and danger to civilians.

Logistical Access:
Despite being isolated, the sites remain within close proximity to major highways and railways to facilitate the transportation of massive rocket components and fuel.

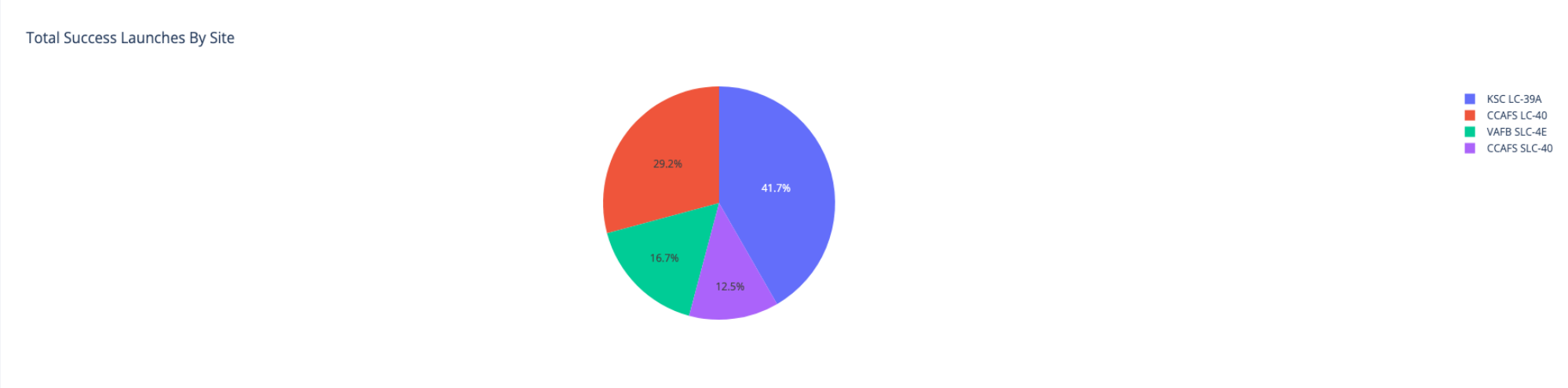
This map illustrates the geographic proximity of the CCAFS SLC-40 (almost same for CCAFS LC-40) launch site to key infrastructure using Folium PolyLines. The distance calculations reveal a strategic operational balance: the site is kept far from densely populated cities like Cape Caneveral (~18.16 km) for safety and acoustic mitigation, while remaining close to the coastline (~0.9 km) to ensure safe over-water flight paths. Additionally, the immediate proximity to railways (~1.5 km) and highways (~18.1 km) allows for efficient logistical transport of massive rocket components and fuel.

Section 4

Build a Dashboard with Plotly Dash

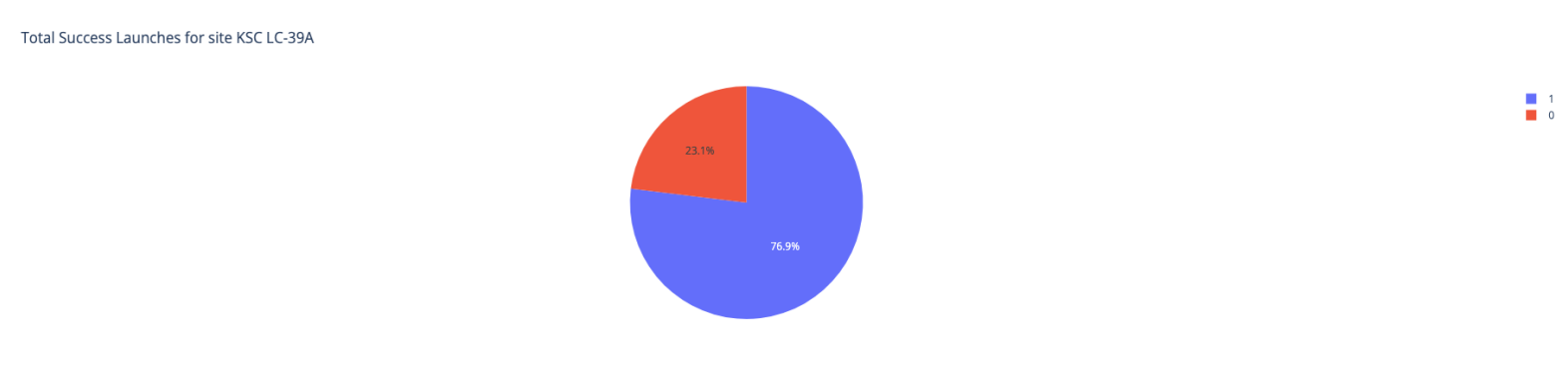
Overall Launch Success Distribution

This pie chart illustrates the total number of successful Falcon 9 first-stage landings distributed across all SpaceX launch facilities. The chart highlights that KSC LC-39A accounts for the largest proportion of total successful recoveries (approximately 41%), followed by CCAFS LC-40. The key finding is that while SpaceX operates multiple sites, the bulk of their successful historical landings in this dataset are heavily concentrated in the Florida-based spaceports.



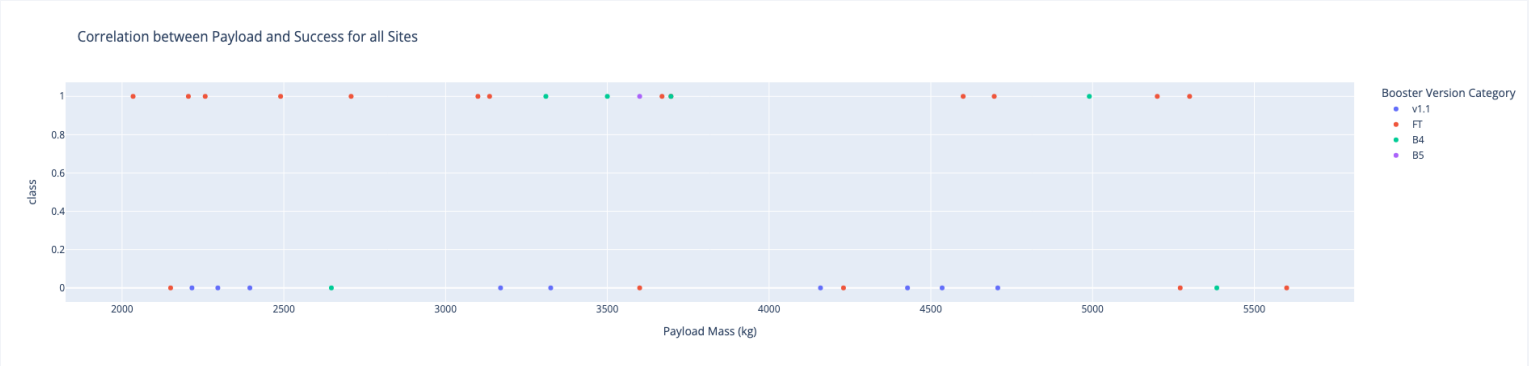
Success Ratio for Top Performing Site (KSC LC-39A)

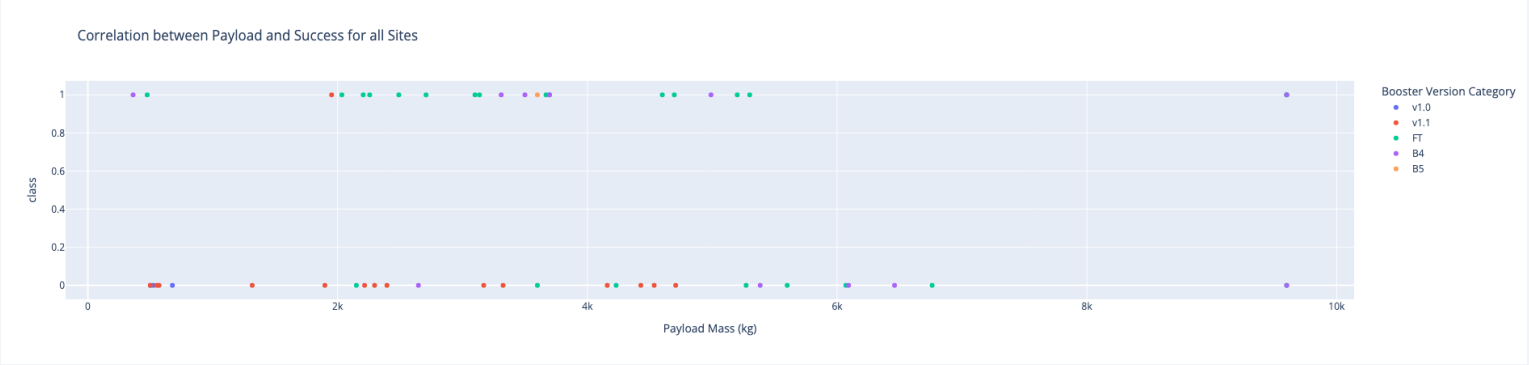
This pie chart details the success-to-failure ratio for KSC LC-39A, which holds the highest launch success rate among all SpaceX facilities in this dataset. The chart clearly illustrates that a significant majority of launches from this site (approximately 77%) resulted in a successful first-stage landing. The key finding is that KSC LC-39A is the most reliable operational site, showing a highly optimized launch and recovery profile compared to the other locations.



Impact of Payload and Booster Version on Landing Outcomes

These scatter plots visualize the correlation between payload mass and landing outcomes across all sites. The first chart displays the entire payload range (0 to 10,000 kg), revealing the overall distribution and showing that extreme weights present more mixed outcomes. The second chart focuses specifically on the 2,000 kg to 6,000 kg range. In this filtered view, a clear 'sweet spot' emerges, showing a dense concentration of successful landings (Class 1). Furthermore, the color-coding indicates that within this optimal weight range, newer booster versions like 'FT' and 'B5' demonstrate exceptional reliability.



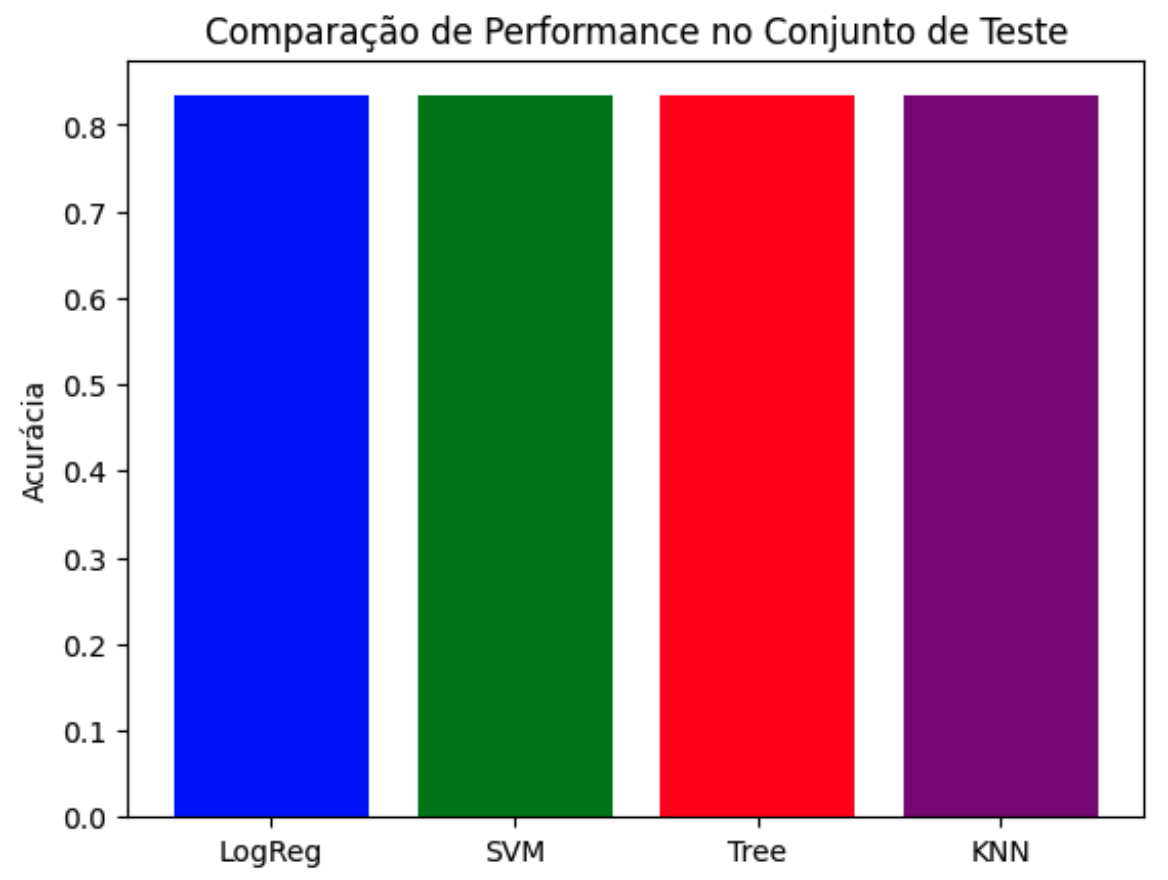


Section 5

Predictive Analysis (Classification)

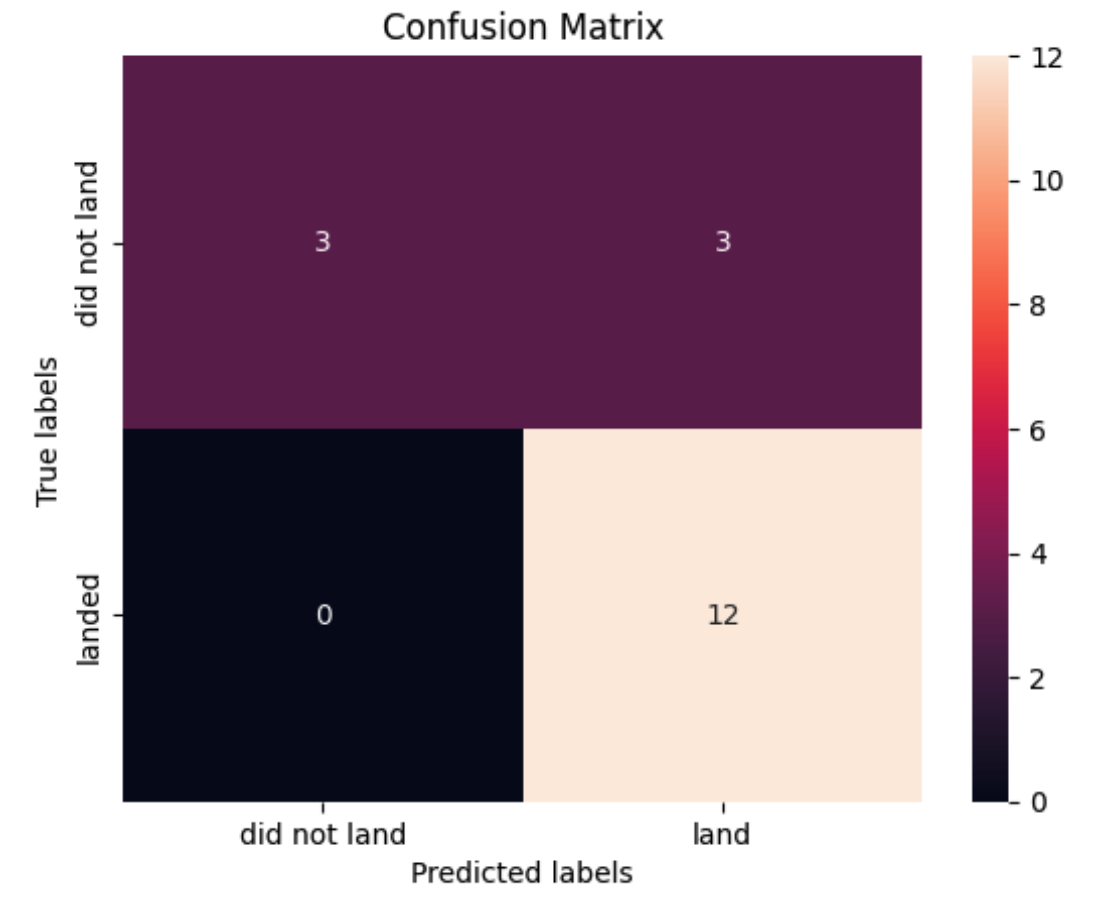
Classification Accuracy

- **Uniform Test Performance:**
  The bar chart demonstrates that all evaluated models (LogReg, SVM, Decision Tree, and KNN) achieved an identical test accuracy of 83.33%.
- **Best Model Selection:**
  Due to the perfect tie in real-world test performance, **Logistic Regression (LogReg)** is highlighted as the optimal model.
- **Conclusion:**
  LogReg is an excellent, lightweight algorithm for this binary classification task. It proves that for this specific dataset, highly complex models do not provide better generalization than a straightforward logistic approach.



Confusion Matrix

- **Consistent Results:** All four classification models produced the exact same confusion matrix, reflecting a shared 83.33% accuracy on the test set.
- **True Positives (12):** The models perfectly predicted all 12 successful landings present in the test data.
- **True Negatives (3):** The models correctly identified 3 failed landings.
- **False Positives (3):** The 16.67% error rate comes exclusively from 3 instances where the model predicted a "Success" (1), but the actual outcome was a "Failure" (0).
- **Finding:** The models are highly effective at identifying success but tend to be slightly over-optimistic, misclassifying a small number of failures as successes.



Conclusions

Predictive Capability:

SpaceX launch outcomes can be predicted with high confidence (83%+) using features like payload mass, orbit type, and launch site.

Model Parity:

Logistic Regression, SVM, KNN, and Decision Trees all performed equally well on the test data, suggesting that the underlying patterns in the SpaceX dataset are robust and consistently recognized by different algorithms.

Key Success Factor:

The "Booster Version" and "Flight Number" are the strongest indicators of success, visually and mathematically proving SpaceX's rapid technological maturation (learning curve).

Strategic Value:

These predictive models can be used to estimate the financial risk and insurance costs of future missions based on historical performance trends.

Appendix

- **Data Sources:** SpaceX REST API and Wikipedia Web Scraping.
- **Tools Used:** Python (Pandas, Numpy, Matplotlib, Seaborn, Folium, Scikit-learn), SQL (SQLite), and Plotly Dash.
- **Methodology:** Complete Data Science pipeline including Data Wrangling, Exploratory Data Analysis (EDA), Interactive Mapping, and Machine Learning Classification.

Thank you!